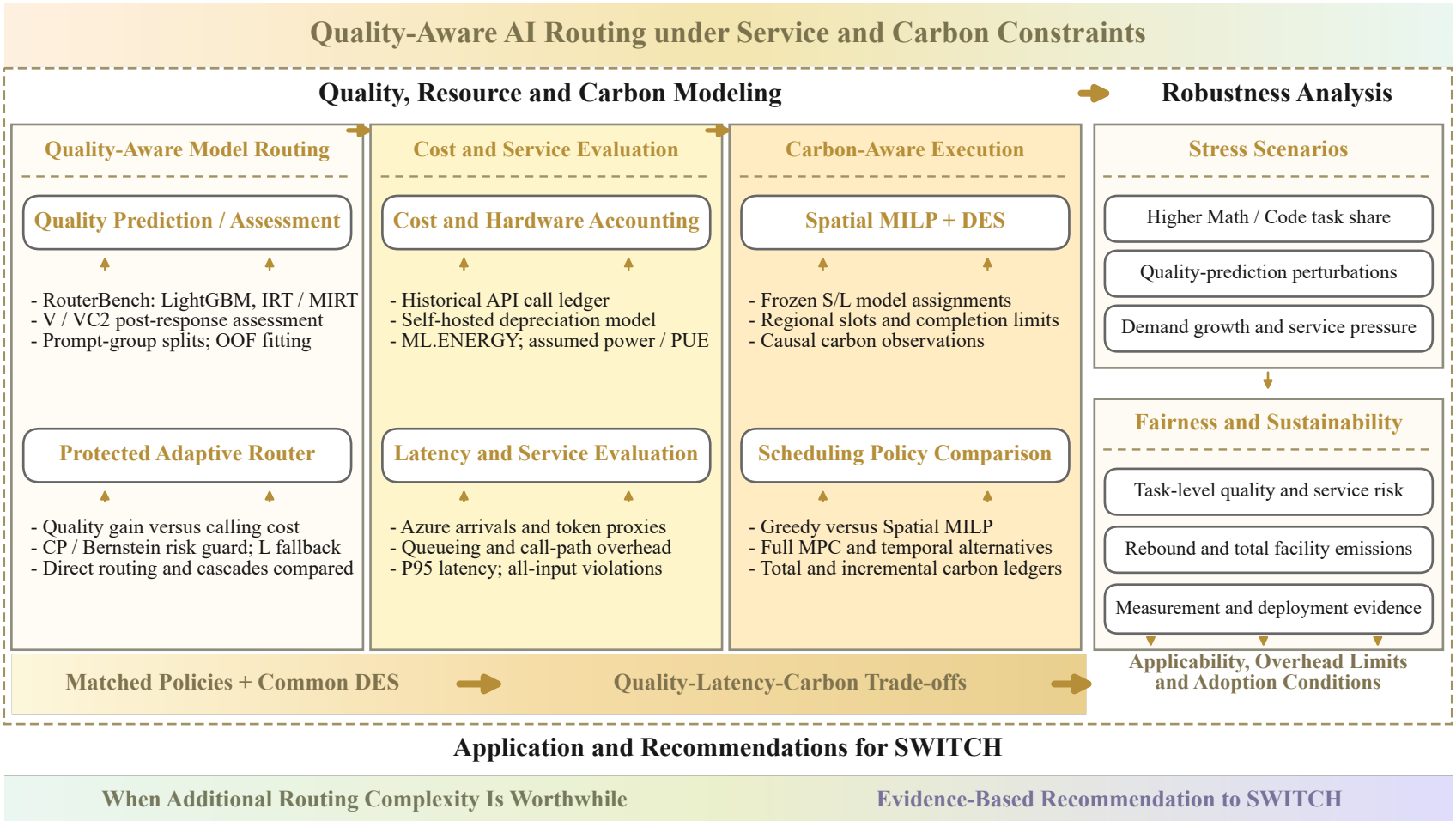

Quality-Aware AI Routing under Service and Carbon Constraints
Quality, Resource and Carbon Modeling
Robustness Analysis
Quality-Aware Model Routing
Quality Prediction / Assessment
- RouterBench: LightGBM, IRT / MIRT
- V / VC2 post-response assessment
- Prompt-group splits; OOF fitting
Protected Adaptive Router
- Quality gain versus calling cost
- CP / Bernstein risk guard; L fallback
- Direct routing and cascades compared
Cost and Service Evaluation
Cost and Hardware Accounting
- Historical API call ledger
- Self-hosted depreciation model
- ML.ENERGY; assumed power / PUE
Latency and Service Evaluation
- Azure arrivals and token proxies
- Queueing and call-path overhead
- P95 latency; all-input violations
Carbon-Aware Execution
Spatial MILP + DES
- Frozen S/L model assignments
- Regional slots and completion limits
- Causal carbon observations
Scheduling Policy Comparison
- Greedy versus Spatial MILP
- Full MPC and temporal alternatives
- Total and incremental carbon ledgers
Stress Scenarios
Higher Math / Code task share
Quality-prediction perturbations
Demand growth and service pressure
Fairness and Sustainability
Task-level quality and service risk
Rebound and total facility emissions
Measurement and deployment evidence
Matched Policies + Common DES
Quality-Latency-Carbon Trade-offs
Applicability, Overhead Limits and Adoption Conditions
Application and Recommendations for SWITCH
When Additional Routing Complexity Is Worthwhile
Evidence-Based Recommendation to SWITCH